

inductance. The situation can be controlled, first by restricting the urge to have tracks that change layers, second by optimising the size of anti-pads ($< 1\text{mm}$) and third by using microvia technology, where anti-pads are largely eliminated.

Optimising anti-pad design

Another issue with dense boards is that anti-pads should not merge with one another, otherwise these can create a larger opening that obviously hinders free flow of return currents and ruins the performance of the plane. In such cases, anti-pads' dimensions and placement must be such that there is a web of 0V between these.

Connection of devices to planes

High contact impedance caused by improper connections of devices to 0V or power planes can be enough to compromise the low impedance benefits gained by using planes for 0V and power. Low inductance can be ensured by the following considerations:

1. Reducing the length and increasing width of the traces between pads and via connecting the component to 0V plane. The best way is to eliminate such traces altogether by using via-in-pad design.

2. Reducing the spacing between the 0V plane and the component, which will reduce the length of the via connecting the component to the plane, thus reducing via inductance.

3. Keeping vias carrying opposite currents as close to each other as possible to maximise their mutual inductance.

Position of devices

Digital ICs and crystals are known to generate intense near-fields in their immediate vicinity that can radiate, causing common-mode noise generation and coupling to nearby traces. This spread can be reduced by using an unbroken ground plane on the immediate bottom layer of

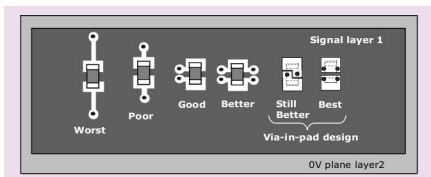


Fig. 12: Connecting devices to planes

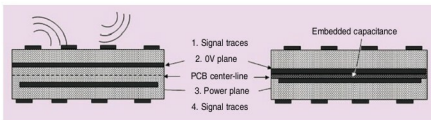


Fig. 13: Microstrip configuration

the noisy component, which serves as an image plane terminating the radiated fields.

If the ICs are located near the edge of the PCB, 0V plane will be only on one side of the IC and the fields can radiate from the other side. Hence, noisy ICs and crystals should never be located on the edge of the PCB. As a general rule of thumb, ICs carrying digital signals with rise time of two nanoseconds or analogue components with frequencies more than 200MHz, should have an unbroken plane that extends at least 5mm beyond its periphery.

Layer stacking in boards

Layer stackup in a PCB is an important factor in determining the EMC performance of a product. While deciding the order in which various layers are stacked, following objectives must be kept in mind:

1. A signal layer should always be adjacent to a plane and be tightly coupled (close) to it. This reduces the loop area enclosed by forward and return signals.

2. Power and ground planes should be closely coupled together to maximise decoupling capacitance

and to reduce ground noise.

3. High-speed signals should be routed on buried layers located between planes. In this way, the planes can act as shields and contain the radiation from high-speed traces.

4. Multiple ground planes are very advantageous, since these will lower the ground (reference plane) impedance of the board and reduce common-mode radiation.

All the above objectives can be met if one uses an eight-layer board. On four- and six-layer boards, objectives 2 and 3 are not achievable and a compromise has to be reached, although multiple ground planes can be provided on a six-layer board, at the expense of a signal layer.

Board stacking for four-layer boards

There are two conflicting considerations while deciding board stacking of four-layer boards. First is the separation between power and ground planes. The lower the separation, the higher is the decoupling capacitance, which can reduce the problem of supply transients and the associated generation of EMI.

Second is the location of signal traces. Keeping power and ground